

STATISTICS

PART 2: Measures of Central Tendency, Measures of Variability

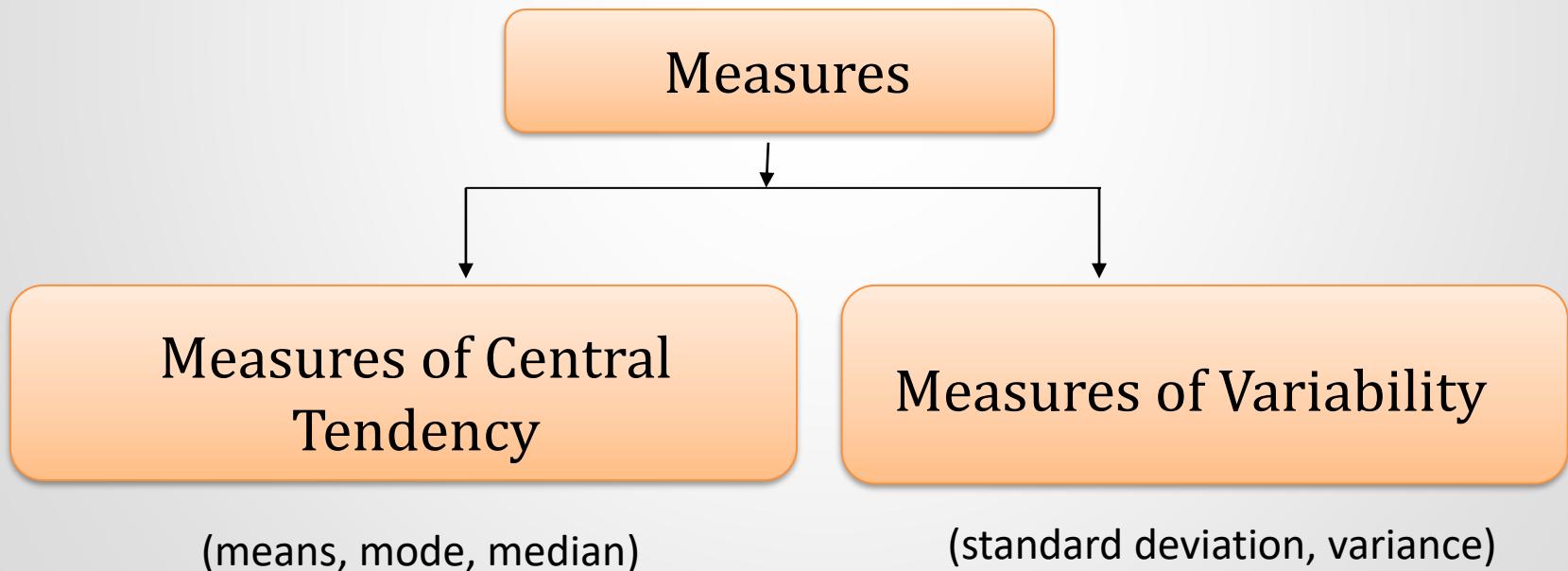
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Numerical Measures

Statistics

A data set can be described (summarized) by using graphics and tables as seen before. A more efficient way of summarizing the data is using numerical measures.



Measures of Central Tendency

Statistics

Measures of Central Tendency

A) Means:

1) Arithmetic Mean: $\bar{x}_a = \frac{1}{n} \sum_{i=1}^n x_i$

$$\bar{x}_a = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i}$$

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

If data are given by frequency distribution (or it is classified)...

$$\bar{x}_a = \frac{2 \times 8 + 3 \times 3 + 4 \times 1}{8 + 3 + 1} = \frac{29}{12} \cong 2.417$$

Measures of Central Tendency

Statistics

Measures of Central Tendency

A) Means:

2) Geometric Mean:

$$\bar{x}_g = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

$$\bar{x}_g = \left(\prod_{i=1}^k x_i^{f_i} \right)^{\frac{1}{\sum_{i=1}^k f_i}}$$



If data are given by frequency distribution (or it is classified)...

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

$$\bar{x}_g = \sqrt[8+3+1]{2^8 3^3 4^1} \cong 2.345$$

Measures of Central Tendency

Statistics

Measures of Central Tendency

A) Means:

3) Harmonic Mean:

$$\overline{x_h} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

$$\overline{x_h} = \frac{\sum_{i=1}^k f_i}{\sum_{i=1}^k \frac{f_i}{x_i}}$$



If data are given by frequency distribution (or it is classified)...

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

$$\overline{x_h} = \frac{8 + 3 + 1}{\frac{8}{2} + \frac{3}{3} + \frac{1}{4}} \cong 2.286$$

Measures of Central Tendency

Statistics

Measures of Central Tendency

A) Means:

4) Quadratic Mean:

$$\bar{x}_q = \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right)^{1/2}$$

$$\bar{x}_q = \left(\frac{\sum_{i=1}^k f_i x_i^2}{\sum_{i=1}^k f_i} \right)^{1/2}$$

If data are given by frequency distribution (or it is classified)...

Example:

Midpoints (x_i)	Frequency (f_i)
2	8
3	3
4	1

$$\bar{x}_h = \sqrt{\frac{8 \times 2^2 + 3 \times 3^2 + 1 \times 4^2}{8 + 3 + 1}} \cong 2.5$$

Measures of Central Tendency

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Measures of Central Tendency

A) Means:

For any data set, the means satisfy the following relation:

$$\overline{x_h} \leq \overline{x_g} \leq \overline{x_a} \leq \overline{x_q}$$

Measures of Central Tendency

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Measures of Central Tendency

B) Median (Me):

To determine **median**, firstly put the numbers in order. Then,

If n is odd, the median is the middle number.

If n is even, the median is the average of two middle numbers.

Example:

$$X = \{2, 3, 1, 9, 2, 3, 4\}$$

$$1, 2, 2, \mathbf{3}, 3, 4, 9$$

$$Me = 3$$

Example:

$$X = \{7, 16, 40, 35\}$$

$$7, \mathbf{16}, \mathbf{35}, 40$$

$$Me = \frac{16+35}{2} = 25.5$$

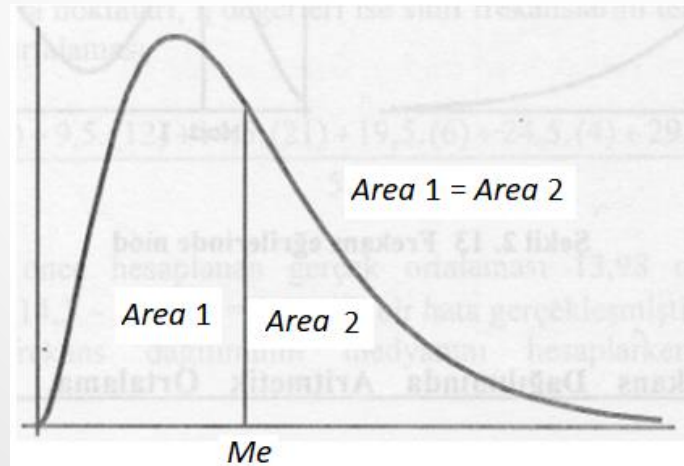
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B) Median (Me):

Consider the following frequency curve.



The median divides the data into two equal parts.

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B) Median (Me):

Median for the classified frequency distribution is computed as:

$$Me = L + \left(\frac{n/2 - (\sum f)_2}{f_{median}} \right) c$$

L : The lower bound of the median class (the class containing the median)

n : The number of the data

$(\sum f)_2$: The sum of the frequencies of the classes that are smaller than the median class

f_{median} : The frequency of the median class

c : The class interval of the median class

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C) Mode(Mo):

The most frequently occurring number in a data set is called the **mode** of the data.

Examples:

$X = \{2, 3, 3, 3, 4, 4, 5, 5, 5, 5, 6, 7, 8\}$ $\Rightarrow Mo = 5$

$X = \{2, 3, 5, 7, 9, 10\}$ $\Rightarrow$ No Mode

$X = \{2, 3, 3, 3, 4, 4, 5, 5, 5, 6\}$ $\Rightarrow Mo = 3 \text{ and } 5$

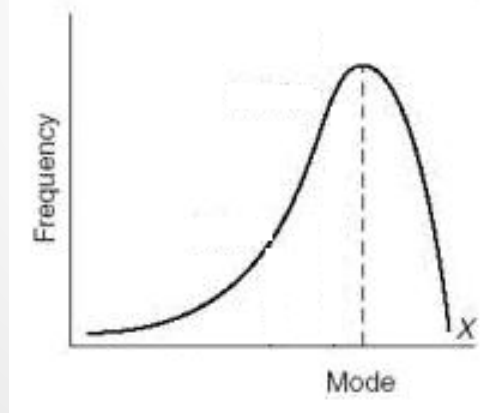
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C) Mode(M_o):

Consider the following frequency curves.



Mode is the point on the x-axis that corresponds to the maximum height of the curve.

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C) Mode(Mo):

Mode for the classified frequency distribution is computed as:

$$Mo = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} c$$

L : The lower bound of the mode class (the class containing the mode)

c : The class interval of the mode class

$\Delta_1 =$

frequency of the
mode class

—

frequency for the previous
class of the mode class

$\Delta_2 =$

frequency of the
mode class

—

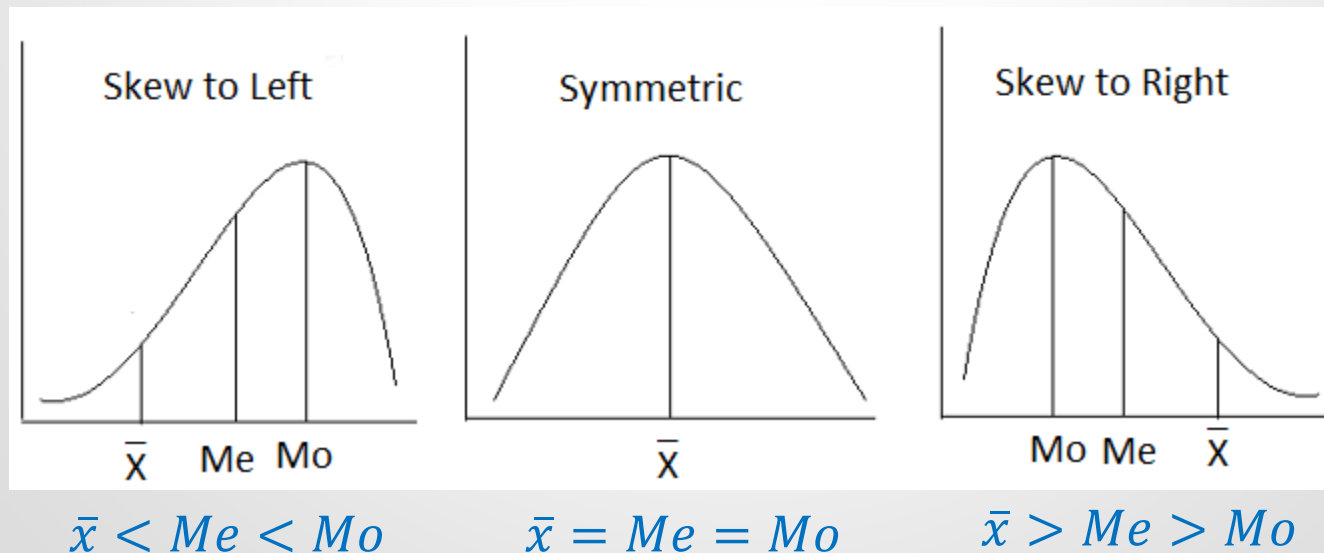
frequency for the next class
of the mode class

Measures of Central Tendency

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Measures of Central Tendency

- Arithmetic Mean → The most important and used measure
- Median → Not sensitive to extreme values compared with arithmetic mean
- Mode → Can be also used for categorical data



Measures of Central Tendency

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Measures of Central Tendency

Example: Consider the **PUZZLE EXAMPLE**.

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2

- Compute arithmetic mean for the exact and classified data. Find the error.
- Compute median for the exact and classified data. Find the error.
- Compute mode for the exact and classified data. Find the error.

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Measures of Central Tendency

Example:

Classes (C_i)	Frequency (f_i)
0-5	1
5-10	2
10-30	3
30-50	4

For the given frequency distribution table, determine the median

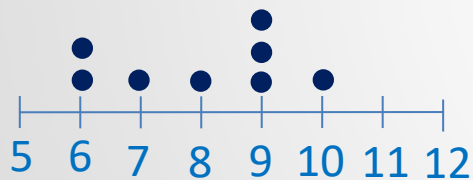
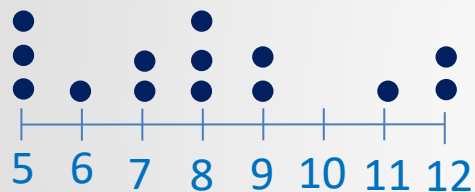
- by using formula.
- by using cumulative frequency polygon.

Measures of Variability

Statistics

Measures of Variability

Consider the following dot plots:



For each distribution, $\bar{x} = 8$. Their mean is the same; however, their distribution (variation) is different as seen clearly. For this reason, we need an extra measure to show this difference.

Measures of Variability

Statistics

Measures of Variability

Consider the data set:

$$X = \{x_1, x_2, x_3, \dots, x_n\}$$

↓ Deviations from the arithmetic mean

$$X = \{x_1 - \bar{x}_a, x_2 - \bar{x}_a, x_3 - \bar{x}_a, \dots, x_n - \bar{x}_a\}$$

If we use the arithmetic mean to summarize this new data...

$$\sigma_a = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_a) = \frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n \bar{x}_a = 0$$

This measure can not be used since it is always 0.

Measures of Variability

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Measures of Variability

The quadratic mean is a more appropriate mean to summarize the given deviations.

$$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

This measure is called **standard deviation**. For the frequency distribution,

$$\sigma_x = \sqrt{\frac{1}{\sum_{i=1}^k f_i} \sum_{i=1}^k f_i (x_i - \bar{x})^2}$$

Unless otherwise stated, we assume that mean refers arithmetic mean, i.e., $\bar{x} = \overline{x_a}$.

Measures of Variability

Statistics

Measures of Variability

The square of the standard deviation, namely σ_x^2 , is called **variance**.

Theorem:

For the exact data,

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2$$

For the frequency distribution,

$$\sigma_x^2 = \frac{1}{\sum_{i=1}^k f_i} \sum_{i=1}^k f_i x_i^2 - \bar{x}^2$$

Prove them!

Measures of Variability

Statistics

Measures of Variability

Example: Consider the **PUZZLE EXAMPLE**. Compute variance for the exact and classified data. Find the error.

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)
2-7	4,5	5
7-12	9,5	12
12-17	14,5	21
17-22	19,5	6
22-27	24,5	4
27-32	29,5	2